

BARUUNKHARAA, Mongolia

WMO: 442410

Lat: 48.9133N

Lon: 106.0897E

Elev: 2648

StdP: 13.34

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.3	-30.0	-38.5	0.7	-33.8	-34.1	0.9	-29.6	17.6	6.3	14.8	2.7	2.1	180	0.607

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.7	90.7	65.4	86.9	64.5	83.4	63.4	70.7	83.9	68.1	80.7	66.0	77.6	5.8	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.5	107.7	76.8	63.9	98.0	73.8	61.9	91.2	71.8	36.5	84.5	34.1	80.9	32.4	77.6	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	16.4	14.0	-36.6	97.9	6.1	4.6	-41.0	101.3	-44.5	104.0	-47.9	106.6	-52.4	109.9		
(5)			-36.7	73.1	6.1	4.0	-41.1	75.9	-44.6	78.2	-48.0	80.4	-52.4	83.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.1	-11.8	-2.2	21.0	40.2	52.5	64.6	69.5	65.1	53.0	35.4	12.9	-5.1		
	DBStd	29.72	9.95	11.93	12.75	9.21	9.21	7.94	6.19	6.67	8.54	9.03	13.00	10.50		
	HDD50	8016	1917	1460	900	315	80	2	0	65	455	1112	1710			
	HDD65	12001	2382	1880	1364	745	399	103	24	82	367	918	1562	2175		
	CDD50	1852	0	0	0	21	159	440	605	469	155	3	0	0		
	CDD65	362	0	0	0	0	13	92	165	85	7	0	0	0		
	CDH74	4333	0	0	0	13	318	1232	1731	901	138	0	0	0		
	CDH80	1572	0	0	0	1	92	470	674	310	25	0	0	0		
	WSAvg	4.6	3.1	3.6	5.1	6.2	6.3	5.6	4.5	4.6	5.0	4.4	4.0	3.4		
	PrecAvg	11.2	0.1	0.1	0.2	0.4	0.9	2.0	2.8	2.6	1.3	0.4	0.2	0.2		
(15) Precipitation	PrecMax	18.6	1.0	0.5	1.1	1.2	2.8	6.1	6.1	6.3	2.7	1.9	0.9	0.8		
	PrecMin	5.3	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.0	0.0	0.0	0.0	0.0		
	PrecStd	3.0	0.2	0.1	0.2	0.3	0.7	1.3	1.4	1.5	0.7	0.4	0.2	0.2		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	19.9	36.3	57.4	75.3	88.7	94.6	97.2	93.6	83.4	67.9	49.5	25.6		
		MCWB	17.5	29.1	42.8	52.1	60.9	64.0	70.6	69.7	62.9	51.0	39.0	22.1		
	2%	DB	12.8	27.7	50.3	68.3	82.1	89.3	91.5	87.3	77.9	61.2	42.5	18.2		
		MCWB	11.0	23.7	39.2	48.8	56.9	63.3	67.0	65.6	59.7	47.0	34.5	16.0		
	5%	DB	7.6	21.4	44.8	63.2	77.1	85.5	87.1	83.1	73.0	56.0	36.2	14.1		
		MCWB	6.3	18.5	35.6	46.1	55.4	62.8	66.3	64.2	57.0	44.2	30.6	12.5		
	10%	DB	3.2	16.2	39.8	58.1	71.9	81.3	83.3	78.9	68.3	51.0	31.4	10.3		
		MCWB	2.1	14.2	32.3	43.5	52.7	61.5	65.2	63.2	54.9	41.3	27.5	9.0		
	0.4%	WB	17.6	29.3	44.1	53.8	64.4	70.9	77.0	73.6	66.0	52.7	39.4	22.3		
		MCDB	19.7	36.1	55.7	71.4	81.7	85.8	90.1	87.2	78.8	65.4	48.3	25.4		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	11.0	23.7	39.6	49.8	59.6	68.0	71.8	69.1	61.8	47.7	34.8	16.2		
		MCDB	12.6	27.6	49.5	66.3	77.8	82.6	84.6	81.4	73.7	59.6	41.8	18.1		
	5%	WB	6.3	18.6	35.9	46.8	56.4	65.6	69.0	66.4	58.5	44.5	31.0	12.5		
		MCDB	7.5	21.4	43.9	62.0	73.9	79.2	81.3	77.2	69.5	55.1	35.7	14.1		
	10%	WB	2.1	14.2	32.4	43.8	53.5	63.3	66.9	64.3	56.0	41.6	27.6	9.0		
		MCDB	3.1	16.2	39.5	57.2	70.4	76.6	78.4	74.5	66.7	50.3	31.4	10.1		
	5% DB	MDBR	18.9	24.6	24.6	25.0	26.4	24.3	21.7	21.9	24.5	24.1	20.5	17.8		
		MCDBR	22.8	29.6	27.9	34.1	35.0	30.6	28.7	28.9	31.2	32.7	25.4	21.4		
		MCWBR	21.2	25.7	18.5	19.5	17.4	12.4	11.4	12.3	16.0	20.0	18.3	19.5		
		MCWBR	22.8	29.5	26.9	31.6	31.9	26.0	23.5	24.2	28.0	31.4	24.6	21.3		
(40) Clear-Sky Solar Irradiance	taub		0.256	0.280	0.323	0.370	0.405	0.412	0.398	0.376	0.327	0.300	0.276	0.259		
		taud	2.268	2.240	2.234	2.206	2.148	2.233	2.327	2.367	2.443	2.447	2.335	2.249		
	Ebn at Noon		257	276	281	277	271	270	272	273	278	268	246	237		
		Edh at Noon	23	30	36	42	46	43	38	35	29	24	21	20		
(44) All-Sky Solar Radiation	RadAvg		574	943	1380	1708	1977	2017	1894	1642	1364	922	578	444		
	RadStd		21	38	63	80	137	113	111	115	83	44	27	26		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (6 neighbors)		N/A	N/A	-3.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page